The Fabius Function is Not an Elementary Function

Density of the analytic locus, and what it costs to prove

Human-readable account of the formal development
Vladimir Reshetnikov's `ProveIt` repository

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Abstract

The Fabius function F is a C^∞ probability distribution function on $[0, 1]$ that is real analytic at no point of that interval. We prove, and formalize in Lean 4 on top of `Mathlib`, that no elementary function of one real variable agrees with F on any subset of $[0, 1]$ with nonempty interior; in particular the restriction of F to $(0, 1)$ is not elementary.

Since F is C^∞, no smoothness obstruction can decide the question: the obstruction must be analyticity. The whole content of the argument is therefore a single positive statement about the class of elementary functions, which we prove in the form we need it and no stronger:

the set of points at which an elementary function is real analytic is a dense open subset of $\mathbb{R}$.

Density is exactly right and cannot be improved: $x \mapsto x^{-1}$ and $x \mapsto |x|$ are elementary and fail to be analytic at the origin. The induction that proves it is short but has one genuinely delicate step, the unary one, where a locally constant intermediate value must be separated from a locally nonconstant one; the separation is furnished by the isolated-zeros theorem for real analytic functions of one variable.

The same conclusion is then proved for a class the inductive definition does not reach: any continuous branch of a polynomial equation whose coefficients are elementary and whose leading coefficient vanishes nowhere. This covers the algebraic functions that are not expressible by radicals, and it rests on a degree induction that avoids the discriminant, of which `Mathlib` has no usable theory, in favour of the analytic implicit function theorem, of which it does.

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1 Statement

1.1 The two inputs

Write $\mathcal{A}(f) = \{x \in \mathbb{R} : f \text{ is real analytic at } x\}$ for the *analytic locus* of $f\colon \mathbb{R} \to \mathbb{R}$. The proof combines one negative fact about the Fabius function with one positive fact about elementary functions.

Theorem 1.1 (Nowhere analyticity; already formalized)**.** *Let F be the Fabius function, extended by 0 to the left of 0 and by 1 to the right of 1. Then $\mathcal{A}(F) = \mathbb{R} \setminus [0,1]$. In particular F is real analytic at no point of $[0,1]$, endpoints included.*

This is the unnumbered corollary to equation (24) of Arias de Reyna, *Arithmetic of the Fabius function* (arXiv:1702.06487v3), transferred from Rvachev's up to F and completed at the right endpoint; in the repository it is `Fabius.canonical_fabius_analyticAt_iff` in the module `FabiusFunction.NowhereAnalytic`. Nothing in the present document reproves it.

Theorem 1.2 (Density of the analytic locus)**.** *If $f\colon \mathbb{R} \to \mathbb{R}$ is elementary, then $\mathcal{A}(f)$ is a dense open subset of $\mathbb{R}$.*

Corollary 1.3 (Main result)**.** *Let $U \subseteq [0,1]$ have nonempty interior and let $g\colon \mathbb{R} \to \mathbb{R}$ be elementary. Then g and F do not agree on U. In particular there is no elementary function that agrees with F on $(0,1)$, and F itself is not elementary.*

Proof. By Theorem 1.2 the set $\mathcal{A}(g)$ is dense, so it meets the nonempty open set $\operatorname{int} U$; pick $x \in \operatorname{int} U \cap \mathcal{A}(g)$. If g and F agreed on U they would have the same germ at x, because $\operatorname{int} U$ is a neighbourhood of x contained in U; analyticity at a point depends only on the germ, so F would be analytic at $x \in [0,1]$, contradicting Theorem 1.1. □

Remark 1.4. Hypothesising nonempty *interior* rather than openness costs nothing in the proof and buys a little: U may be a closed interval, or an interval with some of its rational points removed, or any set that is merely fat somewhere. What cannot be weakened is fatness itself: on a finite set an elementary function certainly can agree with F, since a polynomial interpolates any finite table of values.

Corollary 1.3 is the sharp form of the assertion "the Fabius function is not elementary", and it is strictly stronger than the bare non-elementarity of F as a function on $\mathbb{R}$. The bare statement leaves open the possibility that some elementary function agrees with F on all of $(0,1)$ and differs from it outside; Corollary 1.3 rules that out, and rules it out on every subinterval.

Remark 1.5. Only a very weak consequence of Theorem 1.2 is actually used: that an elementary function is analytic at *at least one* point of every nonempty open set. Density is nevertheless the statement one must prove, because it is the statement that survives the induction: "analytic somewhere" is not preserved by intersection, whereas "dense open" is.

2 The class of elementary functions

2.1 Definition

We follow the description at https://en.wikipedia.org/wiki/Elementary_function: elementary functions are those obtained from polynomials, roots, exponentials, logarithms, and the trigonometric functions and their inverses by finitely many applications of the field operations and composition. Formally:

Definition 2.1 (Elementary functions)**.** The class $\mathcal{E}$ of *elementary functions* $\mathbb{R} \to \mathbb{R}$ is the smallest class of functions such that

1. every constant function $x \mapsto c$ lies in $\mathcal{E}$, and the identity $x \mapsto x$ lies in $\mathcal{E}$;
2. if $f, g \in \mathcal{E}$ then $x \mapsto f(x) + g(x)$, $x \mapsto f(x)g(x)$ and $x \mapsto -f(x)$ lie in $\mathcal{E}$;
3. if $f \in \mathcal{E}$ then $x \mapsto f(x)^{-1}$ lies in $\mathcal{E}$;
4. if $f \in \mathcal{E}$ and $r \in \mathbb{R}$ then $x \mapsto f(x)^r$ lies in $\mathcal{E}$;
5. if $f \in \mathcal{E}$ then each of $\exp \circ f$, $\log \circ f$, $\sin \circ f$, $\cos \circ f$, $\arcsin \circ f$, $\arctan \circ f$ lies in $\mathcal{E}$.

Two features of Definition 2.1 deserve comment.

Composition is a theorem, not an axiom. Definition 2.1 contains no clause "$f, g \in \mathcal{E} \Rightarrow f \circ g \in \mathcal{E}$". It does not need one.

Proposition 2.2. *$\mathcal{E}$ is closed under composition.*

Proof. Induct on the derivation of the outer function f. Every clause of Definition 2.1 commutes with precomposition by a fixed g: the constant $x \mapsto c$ composed with g is again that constant; the identity composed with g is g itself, which is elementary by hypothesis; and $(f_1 + f_2) \circ g = (f_1 \circ g) + (f_2 \circ g)$, $(\exp \circ f) \circ g = \exp \circ (f \circ g)$, and likewise for every remaining clause. ☐

Consequently $\mathcal{E}$ contains every function one expects: differences $f - g = f + (-g)$, quotients $f/g = f \cdot g^{-1}$, natural powers, all polynomial and rational functions, $\sqrt{\cdot}$ and the n-th roots, the absolute value $|x| = \sqrt{x^2}$, the tangent $\sin / \cos$, the hyperbolic functions $\sinh x = (e^x - e^{-x})/2$, $\cosh$, $\tanh$, the arccosine $\pi/2 - \arcsin$, and the inverse hyperbolic sine $\operatorname{arsinh} x = \log\big(x + \sqrt{1 + x^2}\big)$.

Total functions and junk values. Every member of $\mathcal{E}$ is a *total* function $\mathbb{R} \to \mathbb{R}$. This is a deliberate choice, and it is the choice that makes Corollary 1.3 strongest. The formalization inherits `Mathlib`'s conventions

$$0^{-1} = 0, \qquad \log x = \log |x|, \qquad \log 0 = 0, \qquad 0^r = 0 \;\; (r \neq 0), \qquad 0^0 = 1,$$

$$x^r = |x|^r \cos(\pi r) \;\; (x < 0), \qquad \arcsin x = \pm \tfrac{\pi}{2} \;\; (\pm x \geq 1),$$

so that clauses (3)–(5) of Definition 2.1 never fail to produce a function.

One might worry that junk values weaken the theorem, by admitting "elementary" functions that are not classically elementary. They do the opposite. Admitting more functions into $\mathcal{E}$ makes the assertion "$g \in \mathcal{E}$ and $g = F$ on U" easier to satisfy, hence its refutation stronger. And nothing is lost in the other direction: if a classical elementary expression is well formed at every point of an open set U — no vanishing denominator, no nonpositive logarithm, no real power of a nonpositive base, no arcsine of a value outside $[-1, 1]$ — then on U it agrees pointwise with the total function produced by the same derivation, since on U the junk values are never consulted. So every such classical elementary function on U is the restriction to U of a member of $\mathcal{E}$, and Corollary 1.3 applies to it.

Warning 2.3 (Odd roots need a different derivation). The condition "no real power of a nonpositive base" in the box above cannot be dropped, and it is the one place where the total-function convention bites. `Mathlib`'s x^r at a negative base is $|x|^r \cos(\pi r)$, so $(-8)^{1/3} = 2\cos(\pi/3) = 1$, whereas the classical real cube root of -8 is -2. The clause $x \mapsto f(x)^r$ of Definition 2.1 therefore supplies the n-th roots only on $[0, \infty)$.

Nothing is lost, because the classical real odd root is in $\mathcal{E}$ by a *different* derivation,

$$x \longmapsto \frac{x}{|x|} \cdot |x|^{1/n},$$

which uses only division, the absolute value and a real power, and which evaluates to 0 at $x = 0$ under the same conventions. This is `IsElementary.signedRpow`, and `Fabius.signedRoot_pow` verifies formally that for odd n its n-th power is the identity, so it really is the real n-th root and not merely a plausible substitute.

2.2 What is not claimed

$\mathcal{E}$ is closed under n-th roots but not, as defined, under passage to an arbitrary algebraic function: Definition 2.1 has no clause producing a continuous branch of $P(x, y) = 0$ for a polynomial P with elementary coefficients whose Galois group is not solvable, and Liouville's differential-field definition of "elementary" does admit such branches. Section 5 closes that gap, not by enlarging $\mathcal{E}$ but by proving the conclusion of Corollary 1.3 directly for such branches. What remains genuinely outside the result is stated there.

Two smaller gaps are worth naming here.

The first is that $\mathcal{E}$ is a class of *total* functions, so a classical elementary expression enters it only through a derivation whose junk values are never consulted on the region of interest; Warning 2.3 records the one place where the obvious derivation is the wrong one.

The second concerns variable exponents. Definition 2.1 provides $x \mapsto f(x)^r$ for a *fixed* real r, and `IsElementary.rpow_of_pos` supplies f^g whenever the base f is everywhere positive, which covers $x \mapsto x^x$ and every classical variable-exponent expression. The unrestricted two-variable power is not a constructor, because `Mathlib`'s x^y is sign-dependent at a negative base — there $x^y = |x|^y \cos(\pi y)$ — and no single elementary formula covers both signs at once. Nothing is lost for Corollary 1.3, and this is formalized rather than asserted: `Fabius.eqOn_rpow_of_pos` states that on any set where the base is positive, f^g agrees pointwise with $\exp(\log f \cdot g)$, which is a member of $\mathcal{E}$ whenever f and g are. Since Corollary 1.3 already rules out agreement on *every* nonempty open subset of $[0, 1]$, and the base of a classical variable-exponent expression is positive wherever the expression is well formed, such an expression is excluded too.

We also make no claim about elementary functions of a complex variable, and none about the closely related but distinct question of whether F satisfies an algebraic differential equation.

3 Analyticity of the generators

Each generator of Definition 2.1 is real analytic away from a finite set of arguments. All of these facts are in `Mathlib`, most of them in the form "C^n for every n in $\mathbb{N} \cup \{\infty, \omega\}$"; specializing to the analytic exponent ω and applying `ContDiffAt.analyticAt` converts them to the statements below.

Lemma 3.1.

1. $\exp$, $\sin$, $\cos$ *and* $\arctan$ *are real analytic at every point of* $\mathbb{R}$.
2. $\log$ *is real analytic at every* $x \neq 0$ *— at negative arguments as well, because* $\log$ *is even under the convention* $\log x = \log |x|$.

3. *For each fixed* $r \in \mathbb{R}$*, the function* $x \mapsto x^r$ *is real analytic at every* $x \neq 0$*; at negative* x *this uses the identity* $x^r = |x|^r \cos(\pi r)$*.*
4. $\arcsin$ *is real analytic at every* $x \notin \{-1, 1\}$*; outside* $[-1, 1]$ *it is locally constant.*
5. $x \mapsto x^{-1}$ *is real analytic at every* $x \neq 0$*.*

Remark 3.2. The bad sets are sets of *values* of the inner function, not sets of points of the domain — $\{0\}$ for the reciprocal, the logarithm and a real power, $\{-1, 1\}$ for the arcsine, and $\varnothing$ for the rest. This is the shape the induction consumes, and it is why Lemma 4.2 below is stated with a finite set $B \subseteq \mathbb{R}$ of forbidden values.

4 Density of the analytic locus

4.1 Openness

Lemma 4.1. *For every* $f : \mathbb{R} \to \mathbb{R}$ *the set* $\mathcal{A}(f)$ *is open.*

Proof. Analyticity at x means that f agrees near x with the sum of a convergent power series; that series then converges on a whole ball around x and, after a change of origin, represents f near every point of that ball. So $\mathcal{A}(f)$ contains a neighbourhood of each of its points. □

4.2 The unary step

The following lemma is where the argument is actually made. Recall the shape of the difficulty. If f is analytic on a dense open set and g is analytic everywhere except at finitely many values, one would like to say that $g \circ f$ is analytic wherever f is analytic and $f(x)$ avoids the bad values. That set need not be dense: consider $f \equiv 0$ and $g(y) = y^{-1}$. What saves the statement is that in that degenerate case $g \circ f$ is constant, hence analytic anyway. Separating the two cases is exactly the isolated-zeros theorem.

Lemma 4.2 (Key lemma). *Let* $f, g : \mathbb{R} \to \mathbb{R}$ *and let* $B \subseteq \mathbb{R}$ *be finite. Suppose* $\mathcal{A}(f)$ *is dense and* g *is analytic at every* $y \notin B$*. Then* $\mathcal{A}(g \circ f)$ *is dense.*

Proof. Let U be a nonempty open set; we must produce a point of $U \cap \mathcal{A}(g \circ f)$. Since $\mathcal{A}(f)$ is dense there is $x_0 \in U \cap \mathcal{A}(f)$, and by Lemma 4.1 the set $V = U \cap \mathcal{A}(f)$ is an open neighbourhood of x_0. Distinguish two cases.

Case 1: f *is constant equal to some* $c \in B$ *on a neighbourhood of* x_0. Then $g \circ f$ equals the constant $g(c)$ on that neighbourhood, so $g \circ f$ is analytic at x_0 itself, and $x_0 \in U$.

Case 2: for no $c \in B$ *is* f *eventually equal to* c *near* x_0. Fix $c \in B$. The function $f - c$ is analytic at x_0, so by the isolated-zeros theorem for analytic functions of one real variable, either $f - c$ vanishes identically near x_0 — excluded in this case — or $f - c$ is nonvanishing on some punctured neighbourhood of x_0. Thus for each of the finitely many $c \in B$ there is a punctured neighbourhood of x_0 on which $f \neq c$. Intersecting these finitely many punctured neighbourhoods with the punctured neighbourhood $V \setminus \{x_0\}$ leaves a punctured neighbourhood of x_0, which is nonempty because $\mathbb{R}$ has no isolated points. Any point t of it satisfies $t \in U$, f analytic at t, and $f(t) \notin B$; hence g is analytic at $f(t)$ and $g \circ f$ is analytic at t. □

Remark 4.3. The case distinction is not an artefact of the write-up. Without it the lemma is false as stated, for the naive reason above. What makes the two cases exhaustive is the isolated-zeros theorem, applied to $f - c$: an analytic germ either vanishes identically or vanishes nowhere on some punctured neighbourhood. A merely C^∞ inner function admits neither alternative — it can equal c on a sequence accumulating at x_0 without being constant there — so analyticity of f, and not just smoothness, is what the step consumes. This is the only place in the proof where a theorem of substance about real analytic functions is used.

4.3 The induction

Proof of Theorem 1.2. Openness of $\mathcal{A}(f)$ is Lemma 4.1 and holds for every f. Density is proved by induction on the derivation of f in Definition 2.1.

Constants and the identity. Here $\mathcal{A}(f) = \mathbb{R}$.

Sums and products. If $\mathcal{A}(f)$ and $\mathcal{A}(g)$ are dense then so is $\mathcal{A}(f) \cap \mathcal{A}(g)$, because a finite intersection of dense open sets is dense. Analyticity is preserved by sums and products pointwise, so $\mathcal{A}(f) \cap \mathcal{A}(g) \subseteq \mathcal{A}(f+g)$ and likewise for fg; a superset of a dense set is dense.

Negation. $\mathcal{A}(f) \subseteq \mathcal{A}(-f)$.

The unary clauses. Apply Lemma 4.2 with the pair (f, g) and the finite bad set B read off from Lemma 3.1: $B = \varnothing$ for exp, sin, cos, arctan; $B = \{0\}$ for $y \mapsto y^{-1}$, for log, and for $y \mapsto y^r$; and $B = \{-1, 1\}$ for arcsin. □

Remark 4.4 (Sharpness). Density cannot be strengthened to "$\mathcal{A}(f) = \mathbb{R}$": the elementary functions $x \mapsto x^{-1}$, $x \mapsto |x|$ and $x \mapsto \arcsin x$ have analytic loci $\mathbb{R} \setminus \{0\}$, $\mathbb{R} \setminus \{0\}$ and $\mathbb{R} \setminus \{-1, 1\}$. Nor can "dense open" be strengthened to "open with finite complement": the elementary function $x \mapsto (\sin x)^{-1}$ omits the infinite discrete set $\pi\mathbb{Z}$. What density does give for free is that the complement is closed and nowhere dense, being the complement of a dense open set. Whether the complement of the analytic locus of an elementary function is always *discrete* is a natural further question that we neither prove nor use.

5 Algebraic functions over the elementary functions

Definition 2.1 reaches the n-th roots but not a general algebraic function. This section removes that restriction. It does so without enlarging $\mathcal{E}$, for a reason worth stating: an algebraic clause inside the inductive definition would cost Proposition 2.2. Composing a branch y with an elementary h produces a branch over the coefficients $a_i \circ h$, but it is a *continuous* branch only if h is continuous, and elementary functions need not be — $x \mapsto x^{-1}$ is not. Rather than weaken the closure theorem, we prove the conclusion directly.

Theorem 5.1 (Branches of polynomial equations)**.** *Let $U \subseteq \mathbb{R}$ be open, let $a_0, \ldots, a_n$ be real analytic on U with a_n nowhere zero on U, and let y be continuous on U with*

$$\sum_{i=0}^{n} a_i(x)\, y(x)^i \;=\; 0 \qquad (x \in U).$$

Then y is real analytic at some point of every nonempty open subset of U.

Density is again the right conclusion, and for the same kind of reason as before: $y = |\cdot|$ is a continuous branch of $y^2 - x^2 = 0$ on all of $\mathbb{R}$, with $a_0 = -x^2$, $a_1 = 0$, $a_2 = 1$, and it is not analytic at the origin.

Proof architecture. The implicit function theorem applies only where $\partial P/\partial y \neq 0$, and the classical way to control the locus where it vanishes is the discriminant. `Mathlib` has no usable discriminant theory — `Polynomial.discr` carries no bridge to separability, and there is no continuity of roots in the coefficients — so we induct on the degree instead, which needs neither.

Write g for $\partial P/\partial y$ evaluated along the branch and split U into $V = \{x \in U : g(x) \neq 0\}$, which is open, and $W = U \setminus \overline{V}$, which is also open. On V the implicit function theorem applies directly. On W we have $g \equiv 0$, so y satisfies the *derivative* equation, of degree $n-1$, whose leading coefficient $n\, a_n$ is still nowhere zero; the inductive hypothesis applies to it. Finally V together with a dense subset of W is dense in U, because an open set disjoint from V is disjoint from $\overline{V}$ and hence contained in W. The set W absorbs exactly the degenerate locus that a discriminant argument would otherwise have to control, and the induction terminates because the degree drops.

The implicit function step is `Mathlib`'s `ContDiffAt.implicitFunction`, which is stated for every smoothness exponent in $\mathbb{N} \cup \{\infty, \omega\}$; instantiating at ω and converting with `ContDiffAt.analyticAt` gives an analytic implicit function theorem. Its one nontrivial hypothesis, invertibility of the partial derivative as a continuous linear map, is discharged in one variable by `HasDerivAt.hasFDerivAt_equiv` together with uniqueness of the Fréchet derivative. □

Corollary 5.2. *Let $a_0, \ldots, a_n$ be elementary, let a_n vanish nowhere, and let $y \colon \mathbb{R} \to \mathbb{R}$ be continuous with $\sum_{i \leq n} a_i(x) y(x)^i = 0$ for every x. Then y does not agree with the Fabius function on any nonempty open subset of $[0, 1]$; in particular not on $(0, 1)$.*

Proof. Each a_i has dense open analytic locus by Theorem 1.2. The intersection of the $n + 1$ of them is a finite intersection of dense open sets, hence dense and open, and Theorem 5.1 applies to it, giving that the analytic locus of y is dense in $\mathbb{R}$. Now argue as in Corollary 1.3. □

Corollary 5.2 covers every algebraic function over the elementary functions, including those whose Galois group is not solvable and which are therefore not expressible by radicals at all. It is genuinely stronger than Corollary 1.3 in that direction, and independent of it in the other: it says nothing about $\exp$ or $\log$ of an algebraic function, because it is a single algebraic step over elementary coefficients rather than a closure. A tower that interleaves algebraic extensions with exponentials and logarithms at several levels is covered only when it can be presented in one of the two forms.

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6 The formal development

6.1 Correspondence with the Lean names

All statements below live in the namespace `Fabius`.

Object or statement	**Lean name**	**Module**
Elementary functions (Definition 2.1)	`IsElementary`	`ElementaryFunction`
Closure under composition (Proposition 2.2)	`IsElementary.comp`	`ElementaryFunction`
Derived closures: $-$, $/$, x^n, $\sqrt{\ }$, $\lvert\cdot\rvert$, tan, sinh, cosh, tanh, arccos, arsinh, polynomials	`IsElementary.sub`, `.div`, `.npow`, `.sqrt`, `.abs`, `.tan`, `.sinh`, `.cosh`, `.tanh`, `.arccos`, `.arsinh`, `.rpow_of_pos`, `.polynomial`	`ElementaryFunction`
Analytic locus $\mathcal{A}(f)$	`analyticLocus`	`ElementaryFunction`
Openness (Lemma 4.1)	`isOpen_analyticLocus`	`ElementaryFunction`
Generators (Lemma 3.1)	`analyticAt_log_of_ne_zero`, `analyticAt_rpow_const`, `analyticAt_arctan`, `analyticAt_arcsin`; for $x \mapsto x^{-1}$, exp, sin, cos the `Mathlib` lemmas `analyticAt_inv`, `analyticAt_rexp`, `Real.analyticAt_sin`, `Real.analyticAt_cos` are used directly	`ElementaryFunction`
Auxiliary: analyticity of $\sqrt{\cdot}$ off 0 (not used in the induction, which routes $\sqrt{\cdot}$ through `rpow`)	`analyticAt_sqrt_of_ne_zero`	`ElementaryFunction`
Signed root, and its defining property (Warning 2.3)	`IsElementary.signedRpow`, `signedRoot_pow`	`ElementaryFunction`
Variable exponents at a positive base	`IsElementary.rpow_of_pos`, `eqOn_rpow_of_pos`	`ElementaryFunction`
Membership of the named functions	`isElementary_exp`, `_log`, `_sin`, `_cos`, `_tan`, `_arcsin`, `_arccos`, `_arctan`, `_sinh`, `_cosh`, `_tanh`, `_arsinh`, `_sqrt`, `_abs`, `_inv`, `_rpow`, `_id`	`ElementaryFunction`
Key lemma (Lemma 4.2)	`dense_analyticLocus_comp`	`ElementaryFunction`
Density (Theorem 1.2)	`IsElementary.dense_analyticLocus`	`ElementaryFunction`
Analytic at a point of any nonempty open set	`IsElementary.exists_analyticAt_of_isOpen`	`ElementaryFunction`
Exceptional set closed with empty interior	`isClosed_compl_analyticLocus`, `IsElementary.interior_compl_analyticLocus`	`ElementaryFunction`
Analytic implicit function theorem in two real variables	`analyticAt_implicitFunction`, `analyticAt_of_continuous_branch`	`AlgebraicBranch`
Branches of polynomial equations (Theorem 5.1)	`analyticDenseOn_of_algebraic`, `dense_setOf_analyticAt_of_algebraic`, `branchPoly`, `AnalyticDenseOn`	`AlgebraicBranch`
Algebraic functions over $\mathcal{E}$ (Corollary 5.2)	`not_algebraicBranch_eqOn`, `canonical_fabius_not_algebraicBranch_on_Ioo`	`NotElementary`
Nowhere analyticity (Theorem 1.1)	`canonical_fabius_not_analyticAt`, `canonical_fabius_analyticAt_iff`	`NowhereAnalytic`
Main result (Corollary 1.3)	`canonical_fabius_not_isElementary_on_Ioo`	`NotElementary`
Interior form of the obstruction	`IsElementary.not_eqOn_of_interior_nonempty`; `not_isElementary_eqOn_of_interior_nonempty`, `not_algebraicBranch_eqOn_of_interior_nonempty`	`ElementaryFunction`; `NotElementary`
F is not elementary	`canonical_fabius_not_isElementary`	`NotElementary`
General open subset of $[0, 1]$	`not_isElementary_eqOn`	`NotElementary`
Rvachev's up and the signed extension	`rvachev_not_isElementary`, `extendedFabius_not_isElementary`	`NotElementary`

All modules are under `Analysis/FabiusFunction/Lean/FabiusFunction/`.

6.2 Ingredients taken from `Mathlib`

The formalization is short because `Mathlib` already contains the two theorems of substance that the argument consumes.

- `AnalyticAt.eventually_eq_zero_or_eventually_ne_zero` and its two-function form: the isolated-zeros theorem for a function of one variable over a nontrivially normed field, which is what powers Lemma 4.2. It applies verbatim to $f : \mathbb{R} \to \mathbb{R}$.
- `isOpen_analyticAt`: the analytic locus is open, which is Lemma 4.1 verbatim. (The pointwise form, analyticity propagating from a point to a neighbourhood, is `AnalyticAt.eventually_analyticAt`; the development uses the set-level statement.)
- `ContDiffAt.analyticAt` together with the family of `contDiffAt_...` lemmas stated for exponents in $\mathbb{N} \cup \{\infty, \omega\}$: this is how Lemma 3.1 is obtained, including the two facts that are otherwise awkward over $\mathbb{R}$, namely analyticity of $\arctan$ everywhere and of $\arcsin$ off $\{\pm 1\}$.

6.3 Statement of the main theorem in Lean

```
theorem canonical_fabius_not_isElementary_on_Ioo :
    not (exists g : R -> R, IsElementary g and
         Set.EqOn g (fabiusReal fabius) (Set.Ioo 0 1))
```

(rendered here in ASCII; the source uses the usual notation). The development contains no `sorry`, no `axiom` and no `opaque`, and its axiom set is the `Mathlib` default `propext`, `Classical.choice`, `Quot.sound`.

7 Discussion

7.1 Why smoothness cannot decide the question

The Fabius function is C^∞ on $\mathbb{R}$, and its derivatives satisfy the exact self-similar relation $F'(x) = 2F(2x)$ on $[0, 1/2]$. Any proof of non-elementarity must therefore distinguish F from the elementary functions by a property strictly finer than differentiability of every order. Real analyticity is the natural candidate, and Theorem 1.2 says it is a sufficient one. It is also, in a precise sense, the *only* such candidate we know how to extract from Definition 2.1 directly. Nothing in this document rules out a different separating property; what it does show is that analyticity suffices, and that it is available at the modest cost of Lemma 4.2.

7.2 The role of the endpoints

Theorem 1.1 includes the endpoints 0 and 1, where F is flat to infinite order. It is worth recording that Corollary 1.3 does *not* need them. A set that is open in $\mathbb{R}$ and contained in $[0, 1]$ is automatically contained in $(0, 1)$ — an $\mathbb{R}$-open set containing 0 contains negative numbers — so the witness produced by Theorem 1.2 is always an interior point, and the endpoint case of Theorem 1.1 is never invoked. The same is true if one reads "open" in the relative topology of $[0, 1]$: a nonempty relatively open subset still contains a nonempty $\mathbb{R}$-open subinterval of $(0, 1)$.

What the endpoints do buy is exactness. $\mathcal{A}(F)$ is precisely $\mathbb{R} \setminus [0, 1]$: outside the unit interval F is locally constant, hence analytic, so the nowhere-analytic locus is as large as it can be and no larger.

7.3 What a stronger theorem would require

Two natural strengthenings are visible from here.

1. *Interleaved towers.* Replacing $\mathcal{E}$ by the class of functions lying in an *elementary* differential-field extension of the rational functions in Liouville's sense — allowing arbitrary algebraic branches alongside exponentials and logarithms, at any depth — would subsume both Definition 2.1 and Section 5. (The wider *Liouvillian* class also admits antiderivatives, and so contains erf, Ei and Si, which are not elementary; that class is not what is meant here.) Each of the two ingredients is now formalized separately; what is missing is the closure that lets them alternate. The obstruction is the one identified at the head of Section 5: composing an algebraic branch with an elementary function needs the inner function to be continuous. Repairing that needs the fact that a nonconstant real analytic function on an interval has image containing an interval, so that the preimage of a dense open set is again dense — true, classical, and not formalized here.

2. *Differential algebraicity.* A stronger and genuinely different statement is that F satisfies no algebraic differential equation $P\big(x, F, F', \ldots, F^{(n)}\big) = 0$ with polynomial P. Such a statement does not follow from nowhere analyticity — differentially algebraic functions need not be analytic — and is not addressed here.